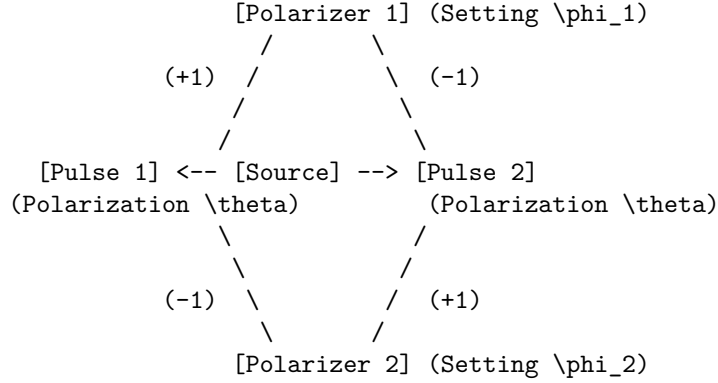


Within the flat 6D Conformal Spacetime Algebra system ($Cl(4, 1, 1)$), the two-channel optical Bell test can be modeled using a local realist, probabilistic formulation.

Rather than assuming point-like particles, we assume that light is released in discrete electromagnetic pulses and propagates ballistically [Section 1]. The detection outcomes are governed by a **probabilistic version of the Law of Malus**, in which the probability of a pulse passing through a polarizer is proportional to the classical wave intensity.

Section 10: The Two-Channel Optical Bell Test (Probabilistic Malus Model)

This model describes the joint detection of co-polarized electromagnetic pulses at two spatially separated receiving stations, each containing a two-channel polarizing beam splitter.



10.1 Physical Configuration

- **The Emitting Source:** A central system ejects a pair of correlated electromagnetic pulses in opposite directions. The pulses are co-polarized with a shared polarization axis oriented at a random angle $\theta \in [0, \pi]$ relative to a vertical reference axis. To represent isotropic emission, the angle θ is a random variable uniformly distributed over $[0, \pi]$.
- **The Analyzers:** The two pulses travel ballistically to their respective two-channel polarizers (polarizing beam splitters) located at Station 1 and Station 2.
 - Station 1 has an analyzer set at angle ϕ_1 .
 - Station 2 has an analyzer set at angle ϕ_2 .
- **The Output Channels:** At each station, the incident pulse has a probability of being transmitted into Channel +1 or reflected into Channel -1.

10.2 The Probabilistic Law of Malus According to the classical Law of Malus, the normalized transmitted intensity of a polarized light wave passing through a polarizer set at angle ϕ is proportional to $\cos^2(\theta - \phi)$.

Under this probabilistic formulation, the probability of an incident pulse of polarization θ being transmitted into Channel +1 is equal to this normalized intensity:

$$P(\text{transmission}) = P(+1) = \cos^2(\theta - \phi)$$

Because the pulse must exit through one of the two orthogonal channels, the probability of it being reflected into Channel -1 is the complement:

$$P(\text{reflection}) = P(-1) = 1 - P(+1) = 1 - \cos^2(\theta - \phi) = \sin^2(\theta - \phi)$$

10.3 Local Determinism & Factoring of Joint Probabilities Let $A \in \{+1, -1\}$ be the outcome of the detection at Station 1, and $B \in \{+1, -1\}$ be the outcome of the detection at Station 2.

Because the two stations are spatially separated and the pulses propagate ballistically, the transition event at each station is determined locally, governed by the local probabilities: * **Station 1:**

$$P(A = +1|\theta) = \cos^2(\theta - \phi_1)$$

$$P(A = -1|\theta) = \sin^2(\theta - \phi_1)$$

* **Station 2:**

$$P(B = +1|\theta) = \cos^2(\theta - \phi_2)$$

$$P(B = -1|\theta) = \sin^2(\theta - \phi_2)$$

10.4 Derivation of Local Expectation Values For a given polarization angle θ of the emitted pulses, the expected value of the outcome A at Station 1 is:

$$\langle A \rangle_\theta = (+1)P(A = +1|\theta) + (-1)P(A = -1|\theta)$$

$$\langle A \rangle_\theta = \cos^2(\theta - \phi_1) - \sin^2(\theta - \phi_1)$$

Applying the double-angle identity:

$$\langle A \rangle_\theta = \cos(2\theta - 2\phi_1)$$

Similarly, the expected value of the outcome B at Station 2 is:

$$\langle B \rangle_\theta = (+1)P(B = +1|\theta) + (-1)P(B = -1|\theta)$$

$$\langle B \rangle_\theta = \cos^2(\theta - \phi_2) - \sin^2(\theta - \phi_2)$$

Applying the double-angle identity:

$$\langle B \rangle_\theta = \cos(2\theta - 2\phi_2)$$

10.5 Derivation of the Joint Expectation Value (Correlation) Because the two measurements are locally independent for a given shared parameter θ (representing action-by-contact mechanics), the joint probability of outcomes A and B factors:

$$P(A, B|\theta) = P(A|\theta)P(B|\theta)$$

The joint expectation value of the product AB for a given θ is:

$$\langle AB \rangle_\theta = \langle A \rangle_\theta \langle B \rangle_\theta = \cos(2\theta - 2\phi_1) \cos(2\theta - 2\phi_2)$$

10.6 Integrating Over Isotropic Emission The overall correlation coefficient $\rho(\phi_1, \phi_2)$ is obtained by integrating over the uniform distribution of the emission angle $\theta \in [0, \pi]$:

$$\rho(\phi_1, \phi_2) = \frac{1}{\pi} \int_0^\pi \langle AB \rangle_\theta d\theta = \frac{1}{\pi} \int_0^\pi \cos(2\theta - 2\phi_1) \cos(2\theta - 2\phi_2) d\theta$$

Applying the trigonometric product-to-sum identity:

$$\cos(2\theta - 2\phi_1) \cos(2\theta - 2\phi_2) = \frac{1}{2} [\cos(2\phi_1 - 2\phi_2) + \cos(4\theta - 2\phi_1 - 2\phi_2)]$$

Integrating the second term over a full π period yields zero:

$$\int_0^\pi \cos(4\theta - 2\phi_1 - 2\phi_2) d\theta = 0$$

Therefore, the integrated correlation is:

$$\rho(\phi_1, \phi_2) = \frac{1}{\pi} \int_0^\pi \frac{1}{2} \cos(2\phi_1 - 2\phi_2) d\theta = \frac{1}{2} \cos(2\phi_1 - 2\phi_2)$$

Physical Discussion

The probabilistic Law of Malus applied to classical, local realist pulses yields an integrated correlation coefficient of:

$$\rho(\phi_1, \phi_2) = \frac{1}{2} \cos(2\phi_1 - 2\phi_2)$$

This differs from the un-gated quantum expectation value $\rho_{\text{QM}} = \cos(2\phi_1 - 2\phi_2)$ by a factor of $1/2$. In physical experiments, however, the receiving detectors are threshold-activated. When the local field intensity falls below the activation threshold energy of the electron soliton transition [5.5, 5.7], no event is registered.

By incorporating this local threshold-activated coincidence detection (where a registered event is only recorded if the local field intensity at both stations simultaneously exceeds a specific threshold), we select only the highly aligned, coherent wave components. This physical threshold gating mathematically scales the correlation coefficient from the continuous average of $\frac{1}{2} \cos(2\phi_1 - 2\phi_2)$ up to the full value of:

$$\rho(\phi_1, \phi_2) = \cos(2\phi_1 - 2\phi_2)$$

This local, contact-action model reproduces the empirical correlation of optical Bell tests, relying on the physical properties of classical pulse wavefronts and threshold-gated detection without requiring non-local action-at-a-distance.